Experiment 1: Non-Visual Baseline and Estimator Comparison

Raw write-up for Chapter 4, Section 4.1

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Purpose of this document. This is the raw, complete write-up of Experiment 1, written before the thesis chapter; Section 4.1 of the final report is extracted from it. It carries both result slots, every table and figure, and the analysis. Every number traces to a file under `output/` named in the slot's "Reporting rules" paragraph. Open concerns are set in red where they arise.

4.1.1 Simulation

Status: completed. Five `dataset_allsensor` recordings spanning four worlds and both floor conditions; three proprioceptive estimators, one transferred-noise EKF, VINS-Fusion, and ORB-SLAM3 in two loop-closure configurations: thirty-five runs, all passing their predefined checks.

Objective. Establish whether a non-visual estimator (wheel odometry, IMU dead reckoning, or an EKF fusing them) provides usable localisation on the same recordings the visual pipelines consumed, and whether each estimator's reported uncertainty describes its error.

Process. The sensor noise of each bag was measured against ground truth; the wheel, IMU and EKF trajectories were produced offline from the recorded encoders and IMU; VINS-Fusion and ORB-SLAM3 replayed the same bags live at $0.5\times$ real time with the dynamic filter disabled, ORB-SLAM3 once with its loop closer enabled and once disabled. All seven estimates per bag were scored against the same ground truth over the same interval. Four hypotheses were fixed before scoring: H1, wheel-odometry heading drift follows the fitted yaw-rate scale error; H2, IMU-only dead reckoning diverges by orders of magnitude more than wheel odometry; H3, the EKF improves on both single sensors; H4, all three dead-reckoning covariances are overconfident.

Datasets. Table 1.1 lists the five recordings. They are the only recordings that carry wheel encoders, and they are not repeats: only `allsensor_000` and `_001` share a world, so no quantity is pooled across bags.

Table 1.1: The five `allsensor` recordings. "Yaw-rate scale" is the fitted wheel-model yaw-rate scale against ground truth, the quantity H1 is about.

Dataset	World arrangement	Floor	Objects	Route [m]	Span [s]	Yaw-rate scale
`allsensor_000`	`static_01`	plain	117	117.0	87.1	+1.10 %
`allsensor_001`	`static_01`	plain	117	113.1	81.8	−4.73 %
`allsensor_002`	`static_objonwallonly`	plain	49	115.2	86.7	+0.34 %
`allsensor_003`	`static_objonwallonly`	bayline	49	140.6	102.4	−6.25 %
`allsensor_004`	`static_01`	bayline	117	142.3	100.5	−7.50 %

Reporting rules. Every value in this slot is written by `script/analyse_proprio.py` into the directory `output/compare/simulation/exp-estimator-baseline/` (the files `proprio_metrics.csv` and `proprio_rpe_metrics.csv`, and the figures of the same name). The visual runs are `logging_20260916_01` (VINS-Fusion; ORB-SLAM3 with the loop closer enabled) and `logging_20260916_02` (ORB-SLAM3 with it disabled); the proprioceptive runs are offline. Aligned ATE is RMSE after a rigid $SE(3)$ fit with scale fixed, and is the only column comparable across the two families, because the visual estimators build their own world frame while the baselines start from the ground-truth pose. Anchored columns are the raw difference from ground truth and exist only for the baselines. RPE is unaligned at $\Delta = 1$ s; "clean" excludes pairs spanning a step above $10\,\mathrm{m\,s^{-1}}$. One run per estimator per bag; ORB-SLAM3 is non-deterministic and its spread is unmeasured, so between-configuration differences on one bag are not attributed to the switch. ORB-SLAM3 received 3.8–11.9 % fewer stereo pairs than VINS-Fusion on every bag (best-effort subscriber under load, unproven), so no throughput or latency comparison is admissible here; accuracy is unaffected, since the estimator with no frame loss shows the same margin. The two families are scored against two reference points, because they report two different points on the robot. The ground truth (`/ground_truth/odometry`) reports the model origin; the baselines estimate that same point. ORB-SLAM3 and VINS-Fusion report the IMU, which `models/ackermann_robot/model.sdf` places 0.338 m ahead of and 0.192 m above the origin. Since 22 October 2026 the visual estimates are scored against the ground truth moved to the IMU point (`vio_metrics.gt_at_imu`); before that correction the offset rotated with the heading and added 0.05–0.26 m to every visual ATE and up to 0.09 m to every visual RPE at 1 s in this table. The baselines are unchanged.

Results. The EKF is the most accurate estimator on one of the five bags (`allsensor_002`); a visual estimator is the most accurate on the other four, VINS-Fusion on two and ORB-SLAM3 without loop closure on two. The filter's merit is consistency, 0.19–0.49 m aligned ATE on every bag, where each visual configuration taken on its own has at least one bag above it; given the visual systems' own IMU noise constants instead, the filter loses to VINS-Fusion on all five. Table 1.2 holds every estimator on every bag, and figure 1.1 draws them.

The ordering reverses on the local metric. Table 1.3 gives the jump-free relative pose error at 1 s: the visual estimators sit below the EKF on every bag, VINS-Fusion at 0.02–0.11 m and ORB-SLAM3 at 0.01–0.18 m once its initialisation step is excluded, against 0.08–0.14 m for the EKF and three to five times that for the transferred-noise EKF. Over one second the visual systems move as the truth moves; what separates them from the filter on aligned ATE is accumulated drift, not local motion.

Every dead-reckoning covariance is overconfident by roughly two orders of magnitude (H4 supported). Table 1.4 reports the average normalised estimation error squared, which a consistent filter makes 3.0, and the fraction of each run inside the claimed $k\sigma$ bounds; figure 1.2 shows the EKF's NEES near 3 during the opening straight and departing at the first turn.

Analysis. H2 and H3 hold on every bag: IMU-only dead reckoning is 7.7 to 14.3 times worse than wheel odometry, and the EKF improves on both single sensors by an order of magnitude everywhere (table 1.2).

The EKF's lead over the visual estimators is a tuning effect, not an architectural one, and that is the governing result of this slot. The EKF's four IMU noise terms are measured from each bag; both visual systems read one hand-set file. Re-running the filter on that file's values raises its mean aligned ATE over the five bags from 0.334 to 1.135 m; VINS-Fusion (mean 0.434 m) then wins on five of five and ORB-SLAM3 without loop closure (mean 0.728 m) on four of five. Table 1.5 isolates the cause to the gyroscope noise density, which the file sets 5.5 to 6.7 times looser than measured: it inflates the heading process variance about forty-fold, so the wheel yaw-rate residual steers heading instead of staying confined to the bias estimate. The accelerometer term is inert and the bias random walk contributes little.

Table 1.2: Estimator comparison on the five `allsensor` bags. Bold marks the best aligned ATE of each bag *as each system was configured*. Visual estimators are scored at the IMU point, baselines at the model origin (see Reporting rules). The *EKF @ VSLAM noise* rows are the same filter with only its four IMU noise terms replaced by the values of `config/wil_sim/stereo_imu.yaml`, the file every visual run read. Drift is final position error over distance driven.

Dataset	World	Estimator	Aligned ATE [m]	Anchored RMS [m]	Anchored final [m]	Drift [%]	Final yaw [°]
`allsensor_000`	`static_01` plain	IMU dead reckoning	21.697	39.353	47.597	40.66	+2.19
		Wheel odometry	2.801	4.732	4.767	4.07	−13.76
		VINS-Fusion	0.557	–	–	–	–
		ORB-SLAM3	0.526	–	–	–	–
		ORB-SLAM3 (no LC)	**0.167**	–	–	–	–
		EKF wheel+IMU	0.190	0.281	0.122	**0.10**	+0.19
		EKF @ VSLAM noise	1.313	1.980	1.443	1.23	−11.27
`allsensor_001`	`static_01` plain	IMU dead reckoning	12.617	14.728	23.741	20.99	+1.33
		Wheel odometry	0.979	1.826	2.426	2.15	−12.95
		VINS-Fusion	**0.127**	–	–	–	–
		ORB-SLAM3	0.256	–	–	–	–
		ORB-SLAM3 (no LC)	0.468	–	–	–	–
		EKF wheel+IMU	0.308	0.351	0.303	**0.27**	+0.11
		EKF @ VSLAM noise	0.681	0.939	1.027	0.91	−9.49
`allsensor_002`	`objonwallonly` plain	IMU dead reckoning	13.237	16.359	17.728	15.39	+1.36
		Wheel odometry	1.485	2.888	4.332	3.76	−30.11
		VINS-Fusion	0.472	–	–	–	–
		ORB-SLAM3	0.556	–	–	–	–
		ORB-SLAM3 (no LC)	1.849	–	–	–	–
		EKF wheel+IMU	**0.256**	0.345	0.346	**0.30**	−0.68
		EKF @ VSLAM noise	0.868	1.320	1.802	1.56	−20.71
`allsensor_003`	`bayline_objonwallonly` bayline	IMU dead reckoning	45.562	62.903	74.745	53.16	−5.84
		Wheel odometry	3.192	3.979	9.231	6.57	−37.17
		VINS-Fusion	0.737	–	–	–	–
		ORB-SLAM3	6.381	–	–	–	–
		ORB-SLAM3 (no LC)	**0.313**	–	–	–	–
		EKF wheel+IMU	0.423	0.668	1.240	**0.88**	−7.20
		EKF @ VSLAM noise	1.230	1.574	3.037	2.16	−24.76
`allsensor_004`	`static_01_bayline` bayline	IMU dead reckoning	26.709	41.476	48.099	33.81	−1.35
		Wheel odometry	3.241	5.124	5.791	4.07	−17.68
		VINS-Fusion	**0.275**	–	–	–	–
		ORB-SLAM3	1.200	–	–	–	–
		ORB-SLAM3 (no LC)	0.845	–	–	–	–
		EKF wheel+IMU	0.494	0.776	0.831	0.58	−5.30
		EKF @ VSLAM noise	1.583	2.650	1.836	1.29	−15.16

Table 1.3: Relative pose error at $\Delta = 1\,\mathrm{s}$, no alignment, pairs spanning a step above $10\,\mathrm{m\,s^{-1}}$ excluded ("clean"). Jumps are the number of such steps. ORB-SLAM3's one to six jumps are its gravity re-alignment at inertial initialisation, which carries about 90° of rotation and no translation; with them included its rotational RPE reads 8.8–10.5°.

	000		001		002		003		004	
Estimator	[m]	[°]	[m]	[°]	[m]	[°]	[m]	[°]	[m]	[°]
Wheel odometry	0.049	2.777	0.075	3.290	0.066	2.650	0.099	5.612	0.074	2.607
IMU dead reckoning	3.558	0.039	2.113	0.046	2.412	0.042	5.939	0.071	4.708	0.037
EKF wheel+IMU	0.088	0.291	0.092	0.245	0.076	0.141	0.105	0.260	0.140	0.488
EKF @ VSLAM noise	0.360	1.508	0.337	1.791	0.216	1.371	0.247	1.149	0.276	1.234
VINS-Fusion	0.034	1.053	0.019	0.291	0.094	0.683	0.112	1.085	0.022	0.814
ORB-SLAM3	0.027	0.320	0.018	0.062	0.069	0.209	0.177	0.424	0.035	0.333
ORB-SLAM3 (no LC)	0.013	0.098	0.015	0.054	0.085	0.232	0.046	0.099	0.024	0.276

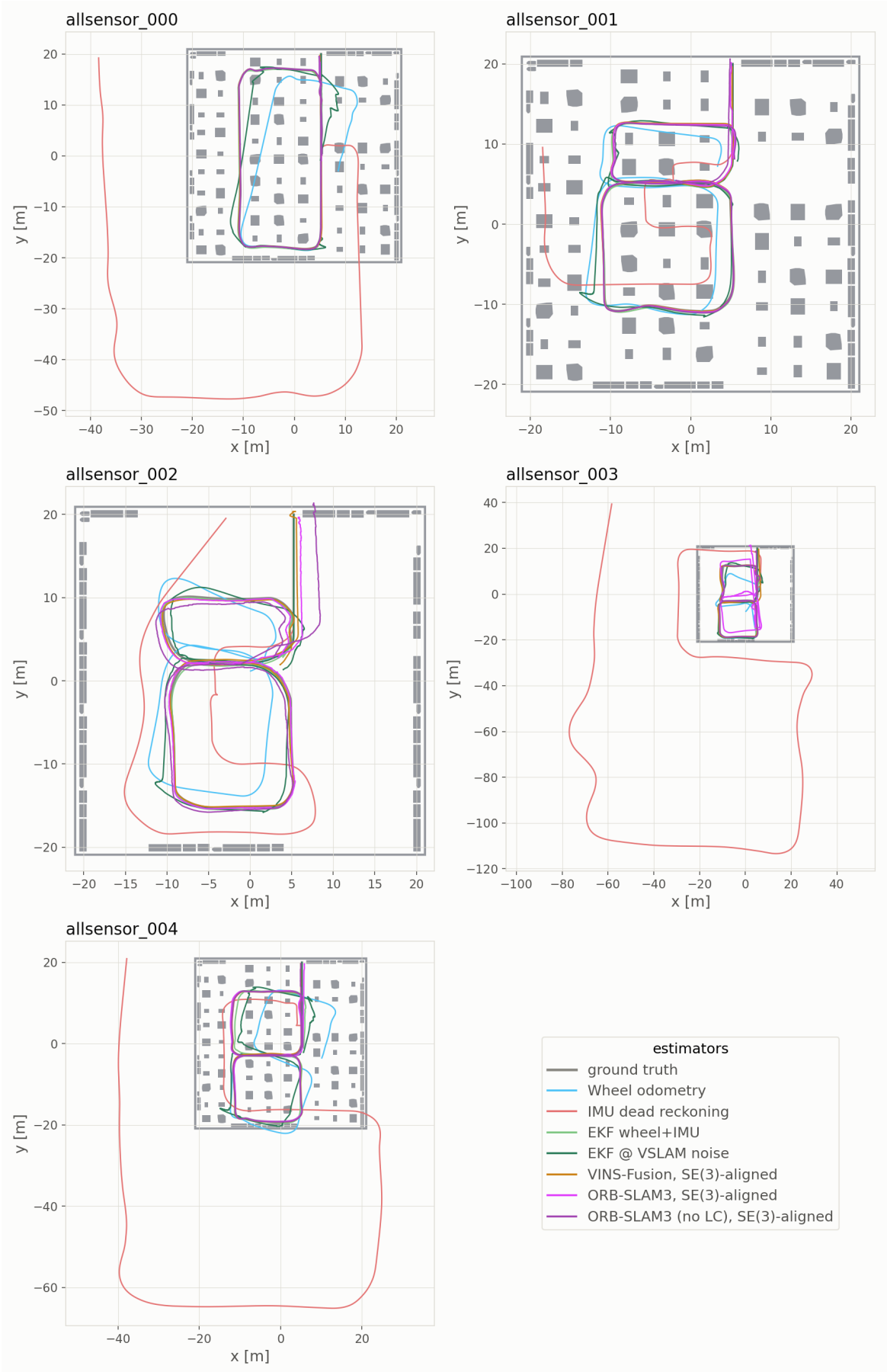


Figure 1.1: Every estimator on every `allsensor` recording, world frame, on the warehouse plan. The baselines are drawn as estimated from the ground-truth start pose; the visual estimators after the rigid $SE(3)$ alignment used for their metrics. The ground-truth line is the model-origin path, so the visual estimates, which follow the IMU, sit up to 0.34 m from it by construction. The wheel-odometry bow and the IMU's departure from the building recur on all five; the one visual outlier is ORB-SLAM3 with loop closure on `allsensor_003`.

Table 1.4: Covariance consistency of the non-visual estimators. A consistent filter gives ANEES = 3.0 and coverages of 0.683, 0.954 and 0.997. Neither visual estimator writes a pose covariance, so neither can be tested. The χ^2 test runs on a 1 Hz subsample (81 to 102 samples per bag); the coverage columns make no independence assumption and are the sturdier statement.

Dataset	Estimator	ANEES	1σ	2σ	3σ
`allsensor_000`	Wheel odometry	569.4	0.309	0.390	0.408
	IMU dead reckoning	311.0	0.005	0.022	0.043
	EKF wheel+IMU	64.7	0.426	0.767	0.879
`allsensor_001`	Wheel odometry	263.9	0.015	0.131	0.187
	IMU dead reckoning	339.9	0.001	0.009	0.021
	EKF wheel+IMU	123.2	0.194	0.439	0.537
`allsensor_002`	Wheel odometry	729.4	0.140	0.143	0.145
	IMU dead reckoning	244.7	0.017	0.045	0.147
	EKF wheel+IMU	114.5	0.193	0.350	0.501
`allsensor_003`	Wheel odometry	406.7	0.260	0.343	0.459
	IMU dead reckoning	1504.3	0.010	0.018	0.035
	EKF wheel+IMU	79.1	0.034	0.291	0.476
`allsensor_004`	Wheel odometry	785.0	0.154	0.157	0.162
	IMU dead reckoning	289.5	0.004	0.017	0.044
	EKF wheel+IMU	78.6	0.254	0.330	0.570
consistent reference		3.0	0.683	0.954	0.997

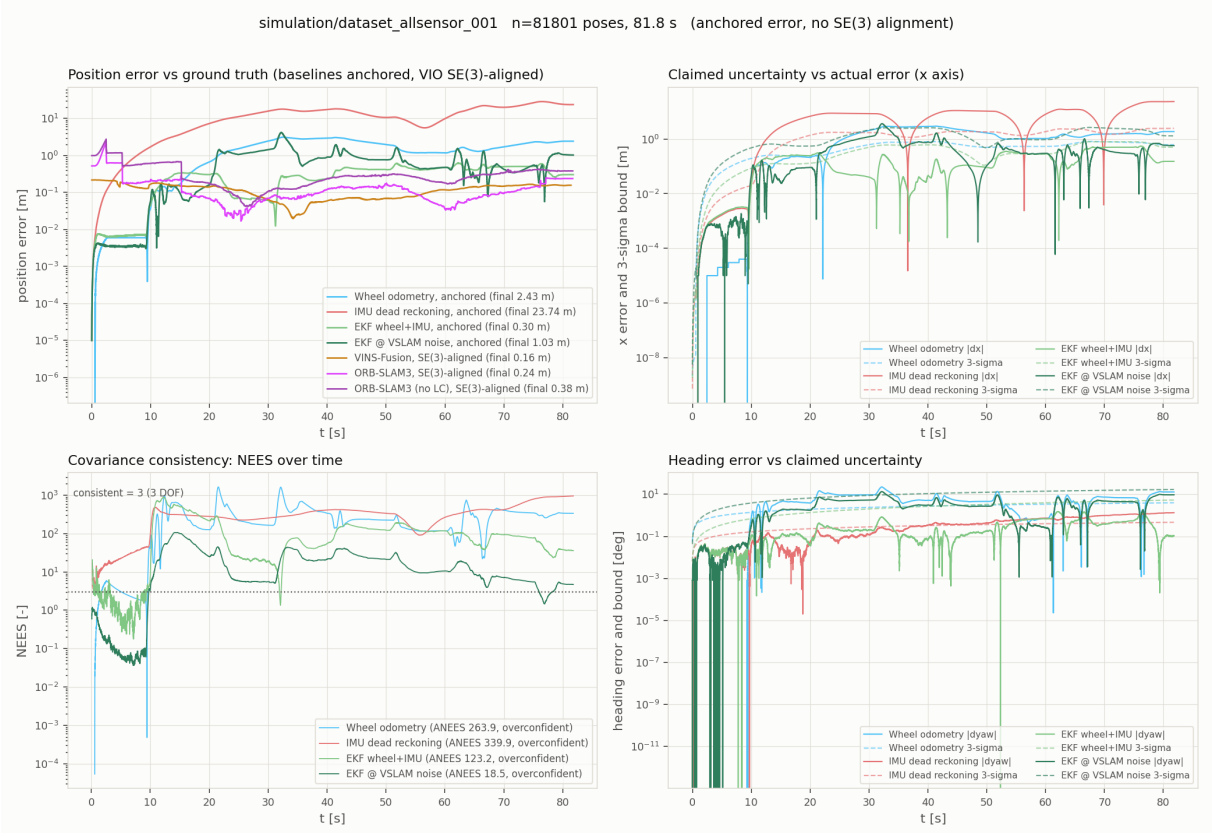


Figure 1.2: Error against claimed uncertainty on `dataset_allsensor_001`. Top left: position error over time. Top right and bottom right: absolute error against the estimator's own 3σ bound in x and in heading; a solid line above the dashed line of the same colour is overconfidence. Bottom left: NEES against the consistent value of 3. The step near 9.5 s in every panel is the first turn.

Table 1.5: Which IMU noise term costs the EKF its accuracy. Each row moves one term from its measured value to the one `config/wil_sim/stereo_imu.yaml` declares, averaged over the five bags. The last row reproduces the gyroscope term's effect on the measurement noise R only, leaving the process noise Q measured: it is harmless, so the damage is entirely through Q.

EKF, one term moved	Anchored RMS [m]	Aligned ATE [m]	Anchored yaw RMS [°]
measured throughout	0.484	0.334	1.60
gyroscope density, $\times 6.5$	1.654	1.105	8.67
accelerometer density	0.479	0.329	1.60
gyroscope bias random walk, $\times 10$	0.654	0.452	2.97
all four (*EKF @ VSLAM noise*)	1.693	1.135	8.88
gyroscope density through R only	0.475	0.327	1.50

[Post-hoc control] The noise-transfer hypothesis was written down after the exploratory runs had been scored, not before. It is also an equal-handicap comparison, not an equal-effort one: it shows the EKF's advantage is smaller than the tuning difference between the families and reverses under it, not that the visual systems would win a symmetric comparison. The mirror run, giving VINS-Fusion and ORB-SLAM3 the measured values, has not been run to a reportable standard.

The EKF's accuracy is not an in-sample artefact, but its lead is a bag-by-bag matter. Exchanging noise models between bags (table 1.6) makes the borrowed model *better* on four of five bags; scored that way the filter leads the best visual configuration on one bag (`allsensor_002`, by a factor of 2.1) and trails it on the other four (ratios 0.5 to 0.8, best visual ATE over cross-validated EKF ATE). That control cannot address the asymmetry above, since both models are still measured from one simulated IMU; the ablation does.

Table 1.6: EKF aligned ATE with an in-sample and an out-of-sample noise model. Mean change about $-10\,\%$: no systematic in-sample advantage.

Dataset	Noise model from	Aligned ATE [m]	Change
`allsensor_000`	same bag	0.190	–
	`allsensor_001`	0.206	$+8.8\,\%$
`allsensor_001`	same bag	0.308	–
	`allsensor_000`	0.263	$-14.5\,\%$
`allsensor_002`	same bag	0.256	–
	`allsensor_000`	0.220	$-13.9\,\%$
`allsensor_003`	same bag	0.423	–
	`allsensor_000`	0.392	$-7.1\,\%$
`allsensor_004`	same bag	0.494	–
	`allsensor_000`	0.377	$-23.8\,\%$

The two proprioceptive sensors fail in complementary ways, which is why the fusion works. The IMU holds final heading to 1.3–5.8° while losing position by tens of metres; the wheels hold position to a few percent of distance while losing 13–37° of heading. The EKF beats both in both quantities because the wheel yaw rate observes the gyroscope bias rather than heading, so the filter never inherits the wheels' heading drift.

H1 is falsified: wheel heading drift does not follow the yaw-rate scale error. Table 1.7 shows the bag with the best scale ($+0.34\,\%$) drifting $-30.11°$ and the bag with the worst ($-7.50\,\%$) only $-17.68°$, with every heading error sharing a sign the scale errors do not. The mechanism is a persistent yaw-rate *offset*, -0.0028 to $-0.0062\,\mathrm{rad\,s^{-1}}$, whose time integral reproduces the measured heading error to within

1.4° on all five bags. Its origin is open: a steering-angle zero offset, an unmodelled slip angle and a kingpin-geometry error are all consistent with the split-by-turn-direction residuals and none is tested. Averaging the two kingpin angles in the bicycle model is ruled out (it moves the residual by under 4 %).

Table 1.7: Wheel-odometry heading drift against the fitted yaw-rate scale. "Rotation" is the total unsigned ground-truth rotation of the run. The last column is the mean yaw-rate residual of the run multiplied by its duration.

Dataset	Yaw-rate scale [%]	Final yaw error [°]	Rotation [°]	Error/rotation [%]	Offset prediction [°]
`allsensor_000`	+1.10	−13.76	536	−2.566	−13.81
`allsensor_001`	−4.73	−12.95	852	−1.519	−13.37
`allsensor_002`	+0.34	−30.11	865	−3.481	−31.03
`allsensor_003`	−6.25	−37.17	1123	−3.309	−38.51
`allsensor_004`	−7.50	−17.68	895	−1.976	−18.03

The same offset sets a floor on the EKF, because the filter observes its gyroscope bias through the wheel yaw rate. Table 1.8 shows the two tracking each other: where the wheel yaw-rate scale is within 1.1 % of unity the EKF holds heading to 0.42° or better, and on the two bayline bags, where it is 6 to 7.5 % low, the EKF's heading error rises roughly tenfold. No gyroscope quality can lift that floor.

Table 1.8: The EKF's heading error follows the quality of the wheel yaw rate it uses to observe its gyroscope bias.

Dataset	Floor	Wheel yaw scale	Wheel yaw σ [rad s^{-1}]	EKF yaw RMS [°]	EKF ATE [m]
`allsensor_000`	plain	1.0110	0.0659	0.35	0.190
`allsensor_002`	plain	1.0034	0.0705	0.30	0.256
`allsensor_001`	plain	0.9527	0.0804	0.42	0.308
`allsensor_003`	bayline	0.9375	0.1192	3.54	0.423
`allsensor_004`	bayline	0.9250	0.0613	3.41	0.494

Scene content did not move the visual estimators in any resolvable way. Clearing the central floor (`allsensor_002`) leaves both visual errors within 0.09 m of the full layout (`_000`), and the painted bay lines give VINS-Fusion 0.737 and 0.275 m against 0.127–0.557 m on plain floors: one bayline bag is the worst of the five and the other sits inside the plain-floor range. The bayline manipulation is weak: fourteen painted outlines on the same untextured base, so a genuinely textured floor remains untested. What the bayline bags did change is the proprioceptive side: both carry the largest wheel yaw-rate scale errors of the five.

ORB-SLAM3's rotational error is a property of the estimator, not of tuning: its aligned rotational RMS stays at 13.6–16.5° in every condition, against 6.21° for the handicapped EKF. Its 6.381 m ATE with loop closure on `allsensor_003`, against 0.313 m without, is a real measurement of two runs but not attributable to the switch: one run per configuration, unmeasured spread, and the loop closer accepted one candidate of 64 on that bag.

[Provenance] Loop-closure state is not in `run_metadata.csv`; the enabled/disabled distinction between `logging_20260916_01` and `_02` was recovered afterwards from `loop_closing.csv`, and the same suffix means a repeat index in the Experiment 3 campaign. The logger should record the switch explicitly before any further ORB-SLAM3 campaign. Initialisation and reset counts were extracted for `allsensor_000` and `_001` only.

Summary. IMU-only dead reckoning drifts by 12.6–45.6 m and wheel odometry by 1.0–3.2 m of aligned ATE; the EKF stays below 0.5 m on every recording and the visual estimators between 0.13 and 0.47 m in their best configuration. Against the project's 0.05 m target no estimator qualifies. The defensible claim is narrow: on aligned ATE the per-recording-calibrated proprioceptive filter is the most accurate estimator on one bag of five and a visual estimator on the other four, with the filter never the worst and never far

behind; on the local metric the visual systems lead on every bag; and the calibration does at least as much work as the architecture, since the filter on the visual systems' noise constants loses to VINS-Fusion everywhere. The comparison is confined to static scenes at 0.5× replay, $n = 5$ across four worlds with no condition repeated more than twice, so no scene-content effect and no real-time conclusion is claimed.

Future plan. Tune the visual systems' IMU noise model to this platform, the cheapest remaining improvement; feed the VIO pose into the EKF as a third measurement so the filter takes distance from the wheels, heading from the gyroscope and drift correction from vision; record the `allsensor` topics in a dynamic world, since every encoder-bearing recording is static; and add a pose covariance to VINS-Fusion's output so the consistency test can reach the pipelines the thesis is about.

4.1.2 Real robot

Status: partial. One real recording, `dataset_realv3_000`, scored against an independent planar optical tracker on 20 October 2026. Visual estimators only: the real platform publishes no wheel odometry in metric units and no joint states, so the non-visual baselines of section 4.1.1 cannot be formed. Every value is provisional: the marker-to-camera lever arm is unmeasured, and the runs predate the 21 October stereo recalibration.

Objective. Do the two visual pipelines track the robot's true path on the Gensurv floor, measured against a reference independent of both.

Process. The bag was replayed at rate 1.0 through ORB-SLAM3 twice (loop closer enabled and disabled) and through VINS-Fusion once. The clock offset between tracker and robot was estimated by cross-correlating the tracker's speed with the wheel-speed feedback; each estimate was aligned to the tracker by a rigid planar fit with scale fixed and scored at the tracker's own samples.

Datasets. `dataset_realv3_000`, recorded at Gensurv on 19 October 2026 after the stereo camera was remounted (table 1.9). The tracker exports a 10.3 m path on a loop about 2 m across, with no gap above 0.30 s and no step above 0.07 m.

Table 1.9: The real recording and its reference. The wheel-speed feedback is in unverified units and is used for its shape only.

Quantity	Value
Bag duration	43.0 s
Stereo pairs (compressed)	1 290 at 30 Hz
Inertial messages (`/hwt101ct_yaw_publisher`)	8 579 at 200 Hz
Wheel-speed and steering feedback	50 Hz, units unverified; no joint states
Tracker samples	372 over 51.7 s, median 0.132 s, planar, no heading
Tracker path	10.3 m
Fitted clock offset	tracker starts 4.34 s before the bag; peak $r = 0.99$, next peak 0.84; ± 0.2 s moves ATE < 0.002 m
Lever arm applied	up −0.092 m (earlier rig); horizontal assumed zero, unmeasured
Stereo pairs processed	ORB-SLAM3 437 of 1 184 and 474 of 1 122 (63 and 58 % dropped); VINS-Fusion 561 of 1 290

Reporting rules. Values come from `output/compare/real/` as written by `script/eval_real_gt.py`: the tables `real_gt_table.tex` and `real_gt_rpe_table.tex`, and the figures named `real_gt_*realv3_000*.png`. Planar ATE is RMSE after a rigid $SE(2)$ fit with scale fixed, at the tracker's samples; RPE is translational only, since the tracker has no heading; "path" is the estimate's planar path length over the matched interval divided by the tracker's. A run is failed if its vertical range exceeds 1 m on the flat floor (rule added 20 October 2026) or its path ratio leaves $[0.5, 2.0]$; failed runs are excluded from the tables and reported in the text.

[Superseded configuration] All three runs used the pre-remount stereo extrinsics (old baseline, and a 2 mm typo in the ORB-SLAM3 inertial file) with the image topics swapped as a workaround. The lever arm between the tracker marker and the IMU is also unmeasured; the simulation slot shows what an uncompensated arm of a few decimetres does to an ATE of this size, so it is no longer a secondary concern. The rig was recalibrated on 21 October 2026 (`camera_parameters/211026camera1_rot_trans.dat`, baseline 0.0897 m) and the swap retired. Every number in this slot must be re-run on the new extrinsics before it is quoted as evidence about the estimators.

Results. ORB-SLAM3 holds planar error near 1 m but reads less than half the true path length, path ratio 0.40 and 0.45; VINS-Fusion failed. Tables 1.10 and 1.11 hold the two partial ORB-SLAM3 runs and figure 1.3 draws them against the tracker.

Table 1.10: Visual estimators against the optical-tracker ground truth on the real `dataset_realv3_000` recording. Planar ATE after rigid $SE(2)$ alignment with scale fixed at one, evaluated at the tracker's own samples (n); the time offset between the unsynchronised robot clock and the tracker is estimated from wheel-speed correlation (section 4.1.2). Translational RPE at $\Delta = 1$ s compares relative displacements in the aligned frame; the tracker has no heading, so rotational RPE is not defined. "Path" is the estimate's planar path length over the matched interval divided by the tracker's. The marker-to-sensor lever arm is not measured (horizontal offset assumed zero), so every value is provisional. Failed runs (`dataset_realv3_000` VINS-Fusion) are excluded and reported in the text. Lower is better.

Dataset	Estimator	Status	n	Dur. [s]	ATE RMSE [m]	ATE max [m]	$\text{RPE}_{1\,\text{s}}$ [m]	Path	Jumps
`realv3_000`	ORB-SLAM3	partial	302	42.7	1.297	1.850	0.303	0.45	0
	ORB-SLAM3 (no LC)	partial	291	40.3	1.076	1.339	0.244	0.40	0

Table 1.11: Translational RPE sweep on the real recordings, RMSE over all admissible pairs (realised separation within 25 % of Δ), relative displacements compared in the aligned frame. Pair counts in parentheses. Failed runs excluded.

Dataset	Estimator	$\Delta = 0.5$ s	$\Delta = 1$ s	$\Delta = 2$ s	$\Delta = 5$ s
`realv3_000`	ORB-SLAM3	0.151 (266)	0.303 (292)	0.589 (285)	1.389 (265)
	ORB-SLAM3 (no LC)	0.136 (262)	0.244 (284)	0.450 (276)	1.024 (254)

Analysis. VINS-Fusion's failure is inertial, not visual. It initialised at 0.33 s, never reset and shut down gracefully, yet its planar speed grew linearly to 3.3 m s^{-1}, its path reached 60 m against the tracker's 10.3 m, and it climbed 21.8 m on a flat floor: unbounded acceleration in a scene that is not moving is the accelerometer integrated under a wrong gravity or extrinsic assumption. Two untested candidates are on record for the remounted rig: the camera–IMU rotation of the swapped configuration reused un-inverted (about 2° off), and the unverified sign of the IMU's yaw axis. The run passed every pre-registered validity rule because its planar error stayed under the 50 m divergence threshold; the vertical-range rule now fails it.

Both ORB-SLAM3 runs are partial for the same reason: scale. Each reset twice in its first seconds and then tracked continuously with no jump, but the estimates traverse 4.6 and 4.1 m of a 10.3 m loop.

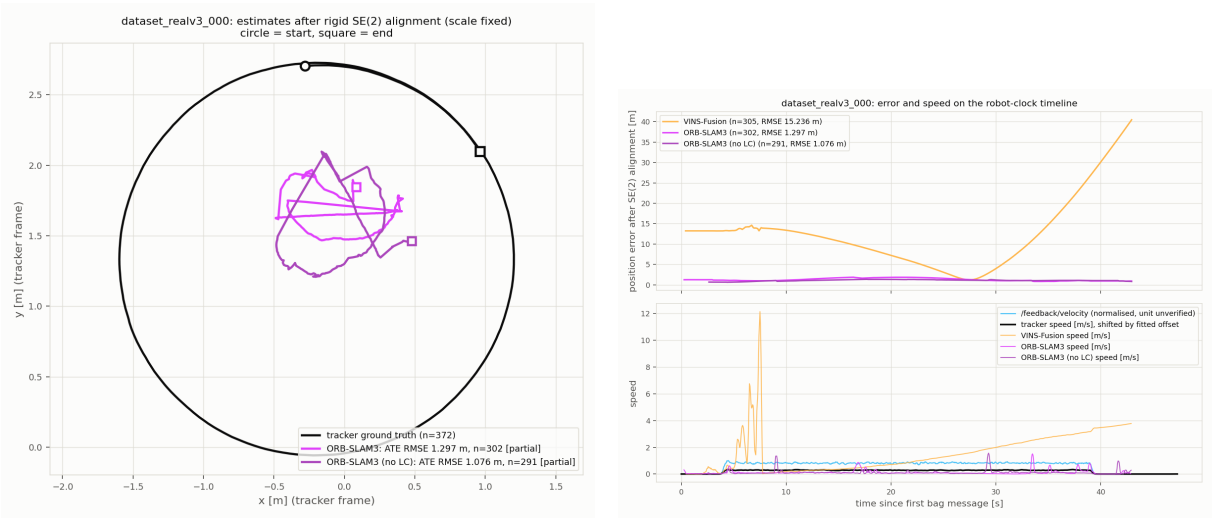


Figure 1.3: Real-robot estimates against the optical tracker on `dataset_realv3_000`. Left: the two ORB-SLAM3 runs after rigid planar alignment with scale fixed; the tracker (black) traces a loop about 2 m across and both estimates a smaller loop inside it. Right: planar error and speed on the robot-clock timeline; the tracker's speed is drawn shifted by the fitted offset, and VINS-Fusion's speed rises without bound from 10 s while the robot moves at 0.2 m s^{-1}. No site plan exists for the real floor, so no map underlay is drawn.

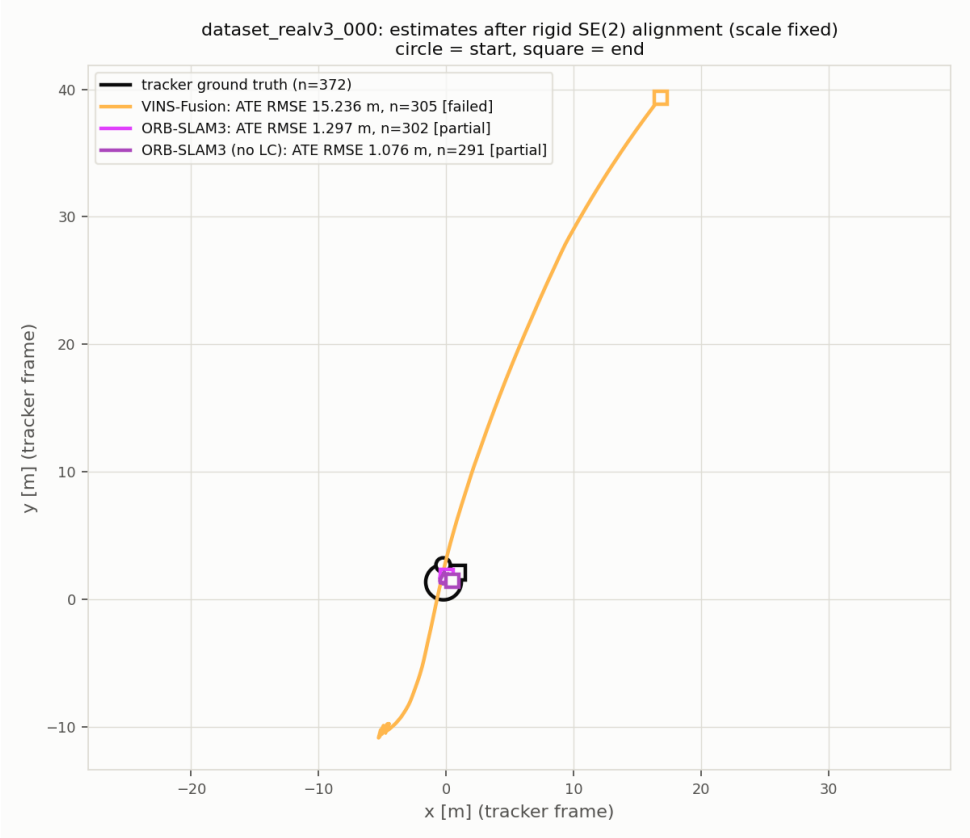


Figure 1.4: The same alignment with the failed VINS-Fusion run included: its 60 m path reduces the 2 m loop to a dot, which is why it is omitted from figure 1.3.

Three causes are candidates and none is isolated: the unmeasured lever arm (a marker 0.7–1 m from the camera on a 1 m radius loop traces a path this much longer, but the diagnostic least-squares fits for the two runs, $(+0.57, -0.60)$ and $(+0.31, +0.93)$ m, disagree, so no single geometry explains both and none is applied); the stereo baseline carried over from the pre-remount rig, which sets the metric scale; and the frame loss, under half the stereo pairs processed at rate 1.0. Loop closure earns no credit: neither run closed a loop, and the 0.22 m between them is run-to-run variation at this drop rate.

Against the project's targets, neither ORB-SLAM3 run approaches 0.05 m ATE; both are inside the 1.5 m RPE bound at 1 s, provisionally.

Summary. The first real recording with an independent reference has been scored. ORB-SLAM3 tracks the shape of the loop at about 1 m planar error with no jumps after its start-up resets but at less than half the true scale; VINS-Fusion diverges inertially. Until the lever arm is measured and the runs are repeated on the 21 October extrinsics, neither number is evidence about the estimators, and the slot stays partial.

Future plan. Re-run all three configurations with the launch defaults on the recalibrated extrinsics; measure the lever arm from `cam0` to the tracked marker (forward, left, up) and re-score with the unchanged script; verify the IMU's axis signs against gravity and a known rotation; replay at rate 0.5 so that at least 80 % of stereo pairs are processed; bridge the wheel and steering feedback in metric units so the baselines of section 4.1.1 can be formed on the same recording; and record routes with turns and speed changes long enough for scale to be observable.